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Systematic Positive EV Extraction from Prop Firms Using Reinforcement Learning

$247 EV per attempt - 2.8x return on deployed capital

Introduction

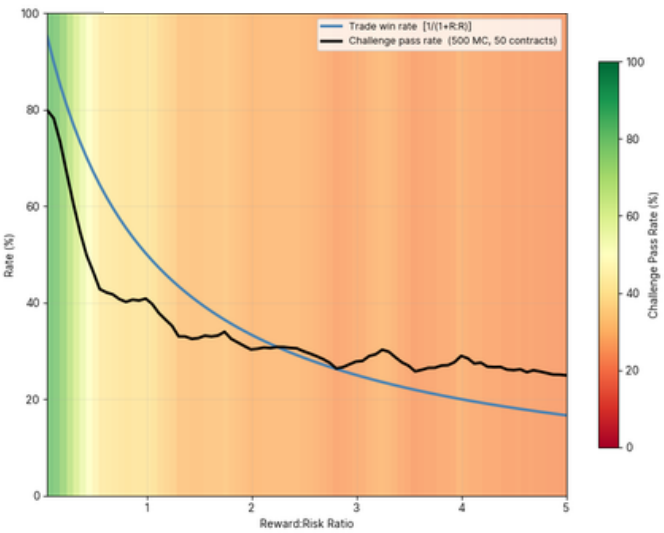
The Thesis

- Prop firms present a unique opportunity within a highly asymmetric risk structure. They effectively create structured products that act like options with a limited downside (49 dollars per month), and infinite upside by access to a funded account (as payouts are not limited)
- This could be hypothetically taken advantage of - you simply build a strategy with two stages to exploit it. The first stage to maximize the chance of passing, and the second stage to maximize the value extraction from the funded account.

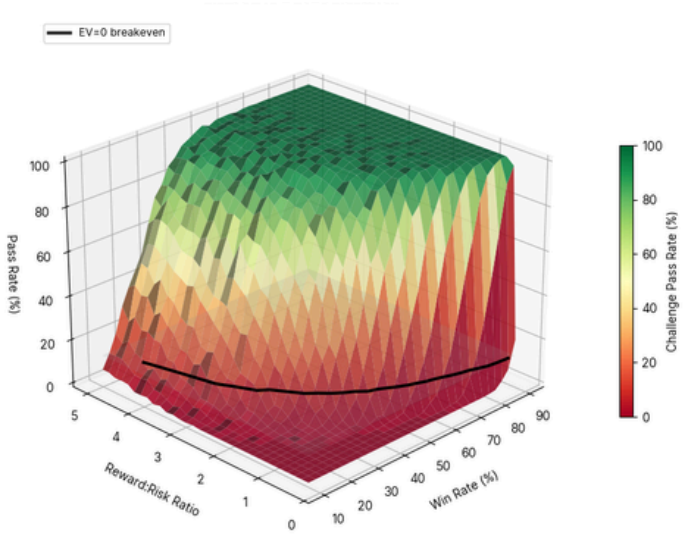
What are prop-firms?

- In this project Topstep's rules are implemented, one of the most popular prop firms.
- On Topstep, you pay a $49 monthly activation fee to attempt a challenge. In this challenge, you are given a 50k balance, and a goal of 6% profit. However, there is a drawdown maximum of 4% - if you hit that, you fail.
- Once this challenge is passed, you are given access to a funded account with a 50k balance again, which you must activate for $149. In this account you keep 90% of your profits, which you can withdraw under certain rules of consistency.

pass rate against risk ratio

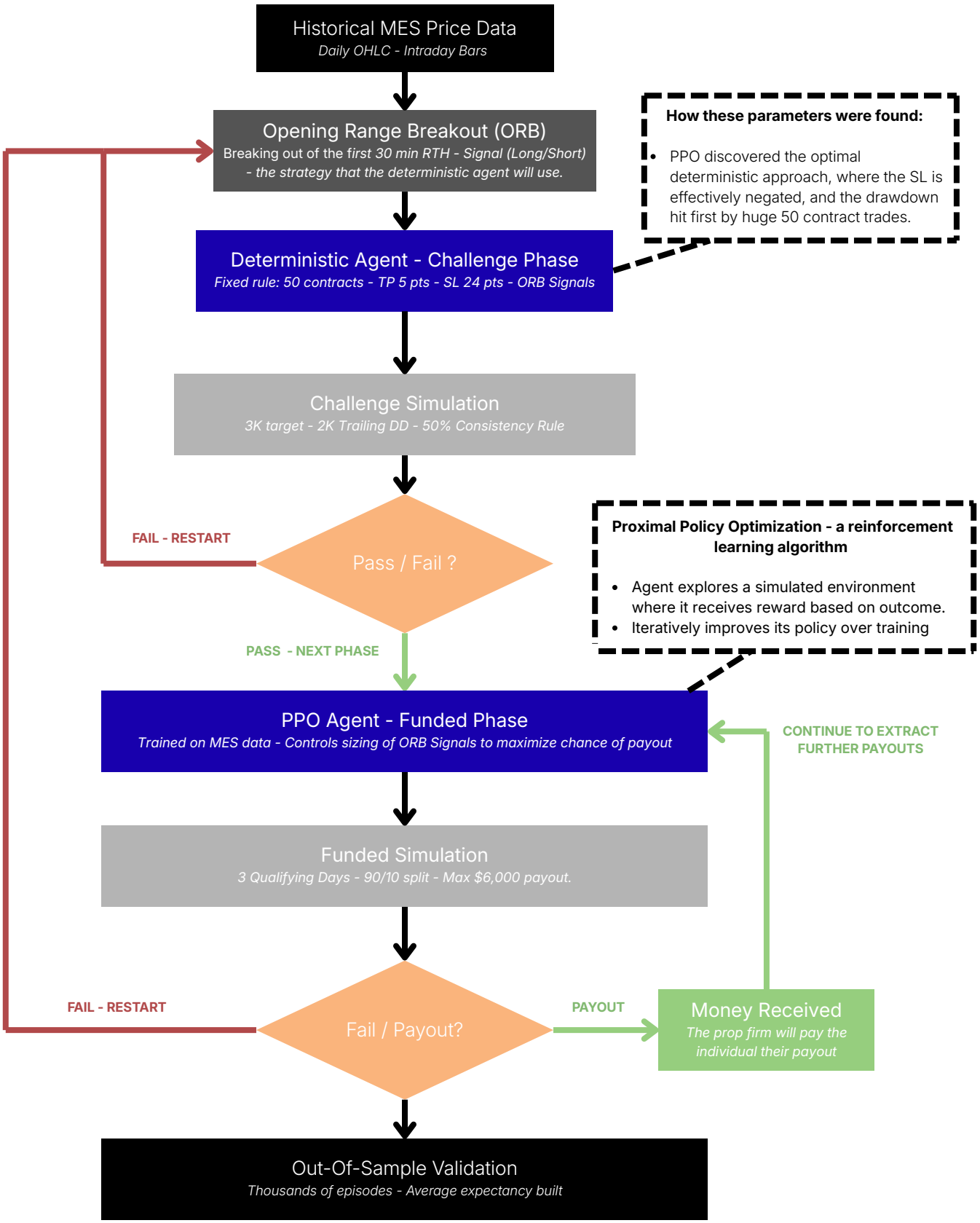


pass rate against the full win rate + risk ratio parameter space



Methodology

All simulations use Databento GLBX.MDP3 1-minute MES bars spanning 2020-01-02 to 2026-03-13 (1,602 trading days), with results reported exclusively on a held-out validation set the model never saw during training.



Data, Implementation, Validation & Limitations

Data

- Source: Databento GLBX.MDP3, 1-minute OHLCV bars, MES continuous futures (MES.FUT). Full dataset: 2020-01-02 to 2026-03-13, 1,602 trading days, 179MB.
- Split chronologically 70/15/15 - train: to 2024-05-02, test: to 2025-04-07, holdout: to 2026-03-13.
- The test split carried indirect selection pressure during training therefore the holdout is the only fully unbiased split, and all headline results are drawn from it exclusively.

Transaction Costs

- Commission: $0.37/leg ($0.74 round-turn per contract).
- Slippage: 0.25pts/leg base, scaling by 0.02 × (n−1) additional contracts, plus 0.25pt additional on breakout entry fills. At maximum size of 50 contracts this totals $86.50 round-turn ($37 commission + $49.50 slippage).
- All reported figures are net of these costs.

PPO Hyperparameters

- Learning rate: 3e-4 — Network: MLP [64, 64] — n_steps: 2048 — batch_size: 256 — n_epochs: 10 — γ: 0.995 — GAE λ: 0.95 — clip: 0.1 — ent_coef: 0.05 — grad norm: 0.5.
- Trained for 4M timesteps across 10 parallel environments, with episodes terminating on payout, blowup, or 200-step limit.
- Dense, per-step rewards provide continuous feedback, with terminal rewards of payout/failure, where payout is significantly more significant than failure, biasing the agent towards profitable sizing without recklessness

Validation

- Headline pass/payout results are from 500 challenge attempts run on the holdout split exclusively (2025-04-08 to 2026-03-13).
- The funded Monte Carlo (2,000 episodes) was run across the combined OOS period — test and holdout together (2024-05-03 to 2026-03-13, 481 days) — as the holdout period alone does not provide sufficient days for a statistically robust Monte Carlo.
- The challenge results remain holdout-only and uncontaminated.

Limitations

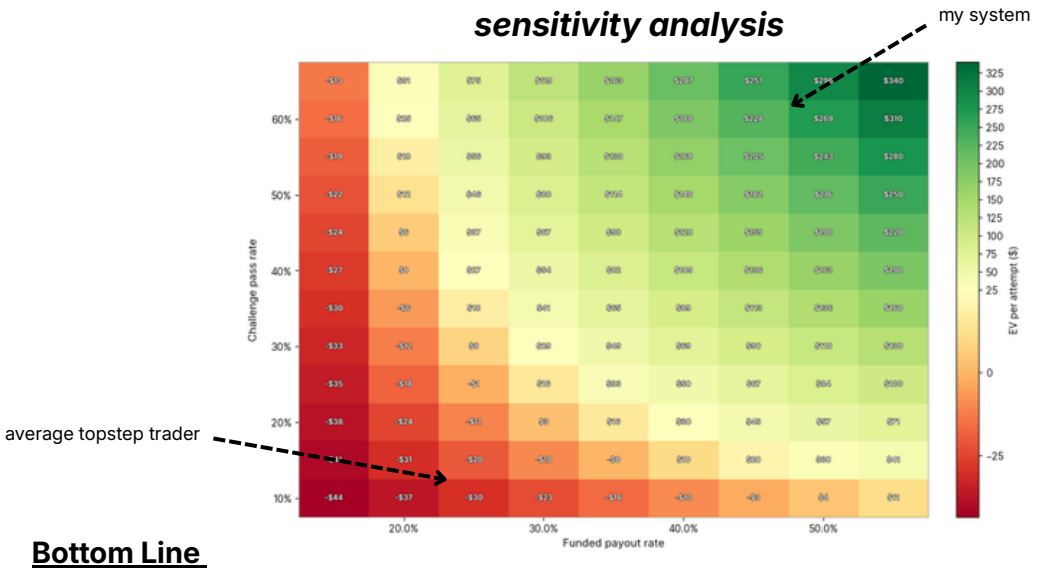
- Platform risk - prop firm rule changes or payout delays and hard ceiling on capital deployment per person per platform constrain scalability. Furthermore, if systematic exploitation of prop firm structures becomes more common, firms may tighten consistency rules or adjust drawdown limits to counter extreme-risk approaches
- Backtest limitations - holdout may not represent future market regimes
- Regime risk - while ORB signal quality may vary across market regimes, the system's EV is robust to moderate signal degradation; the sensitivity analysis shows that value extraction derives primarily from the structural asymmetry of the prop firm rules rather than alpha in the ORB signal.

Economic Analysis

Average Topstep trader statistic based on publicly shared figures

CATEGORY	MY SYSTEM (95% CI)	AVERAGE TOPSTEP TRADER
Monthly Subscription	$49month	$49month
Activation Fee (on pass)	$149 one-time	$149 one-time
Average Pass Rate	58.4% - 95% CI [54.0%, 62.6%]	12.4%
Average Payout Rate	48.3 % - 95% CI [42.6%, 54.0%]	28.2%
Average net payout	$1,358 - 95% CI [$1,318, $1,399]	unknown - assume ours
Expected cost per attempt	$136	$68 (lower due to lower pass rate)
Expected revenue per attempt	$383	$47.49
Expected value per attempt	Revenue - Cost = $247	-$20

sensitivity analysis



Bottom Line

$383 of expected revenue per $136 of expected capital deployed per attempt = 2.8x return, and is highly parallelizable. This strategy may have an even higher EV in reality as the potential for second, third or more payouts has not been included in the economic analysis.